FBA Group Assignment Dataset — Documentation

FBA Teaching Team

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Data for the Group Assignment: Multiple files describing customers, products, marketing campaigns, browsing behaviour and purchases for an online retailer. Eight files in total.

> **The data is not analysis-ready.** It arrives the way data often arrives in practice: spread across several files, with inconsistent column names, some values that cannot be right, and one table stored in an untidy shape. Preparing it is part of the assignment, and your cleaning decisions are assessed.

The business context

Ferdeer Retail is an online retailer selling consumer goods across seven countries. Customers browse the site, add items to a cart, and sometimes buy. Marketing runs campaigns through several channels to bring people to the site and encourage them to purchase.

Over the last three years the company has logged everything: who its customers are, what they browsed, which campaigns they saw, and what they eventually bought. It has also been running an experiment on the site itself, testing whether changes to the checkout process make people more likely to complete a purchase.

The business challenge

Marketing decisions at Ferdeer Retail span the full customer journey: attracting people to the site, understanding how they interact with the business, encouraging them to purchase, and improving the customer experience over time.

Management has identified three areas where it would like to understand the business better:

1. What can we learn about our customers and how they interact with our marketing and purchasing journey?
2. Can the information we have about customers and their visits help us identify who is most likely to make a purchase?
3. When we make changes to the customer experience, do those changes lead to better outcomes?

Each of these is a broad area of interest. Your team will need to decide what, specifically, is worth investigating using the available data.

Management is also open to other ideas. If your team identifies a different question with a similar level of business importance, and you can answer it convincingly using the available data, you are welcome to pursue it instead.

Whatever direction you choose, your analysis should be built around a clear business question and an analytical approach that is appropriate for answering it.

The files (overall description)

File	Rows	Grain (one row per…)
customer_demographics.csv	10,000	customer
customer_profile.csv	10,100	customer (see note)
products.csv	2,000	product
category_lookup.csv	6	product category
campaigns.csv	50	marketing campaign
campaign_spend_wide.csv	50	marketing campaign
events.csv	199,627	user interaction (view/click/cart/bounce/purchase)
transactions.csv	17,800	completed purchase

i Note regarding how the files fit together

Customer information is split across two files. Product categories are stored as codes, with the names in a separate lookup file. Campaign spend is stored with one column per year. You will need to combine these before you can analyse anything.
The tables link on customer, product and campaign identifiers — but **the identifier columns are not named consistently across files**, so check before you join.

customer_demographics.csv

Column	Type	Description
`CustomerID`	int	Unique identifier for the customer. Links to the other customer, event and transaction files.
`Signup Date`	date	Date the customer registered on the platform (2021–2023).
`cntry`	string	Country of residence (US, IN, AU, BR, CA, DE, UK).
`Age`	int	Age of the customer in years. Adult customers only.
`Gender`	string	Male, Female, or Other.

customer_profile.csv

Column	Type	Description
`cust_id`	int	Identifier for the customer. Corresponds to the customer identifier in the other files.
`LoyaltyTier`	string	Loyalty status: Bronze, Silver, Gold, Platinum.
`acquisition channel`	string	How the customer first joined the platform (Organic, Paid Search, Social, Email, Referral).
`ExperimentGroup`	string	Which checkout version the customer was assigned to: **Control**, **Variant_A**, or **Variant_B**. Assignment is random and fixed for each customer.

i Note regarding the site experiment

Every customer has been assigned to one of three versions of the checkout experience, at random, and stays in that version throughout:

- *Control*: The current site, unchanged.
- *Variant_A*: A simplified one-page checkout.
- *Variant_B*: One-click checkout, with a guest option (no account needed).

Both variants aim to reduce friction at the point of purchase. However, the company has not yet evaluated the results.

products.csv

Column	Type	Description
`prod_id`	int	Unique identifier for the product.
`cat_code`	string	Product category, stored as a code. The readable names are in `category_lookup.csv`.
`Brand`	string	Brand associated with the product.
`base price`	float	Base retail price in USD, before any discount. A price should be a positive amount.
`launch_date`	date	Date the product entered the catalogue.
`is_premium`	int (0/1)	1 = premium-priced segment, 0 = standard.

category_lookup.csv

Column	Type	Description
`cat_code`	string	Category code, as used in `products.csv`.
`category_name`	string	Readable category name (Electronics, Fashion, Home, Beauty, Grocery, Sports).

campaigns.csv

Column	Type	Description
`campaign_id`	int	Unique identifier for the campaign.
`Channel`	string	Email, Paid Search, Social, Display, Affiliate.
`objective`	string	Acquisition, Retention, Cross-sell, Reactivation.
`start date`	date	Date the campaign began.
`end date`	date	Date the campaign ended.
`TargetSegment`	string	Segment the campaign was aimed at (New Customers, Churn Risk, Deal Seekers, High Value, All). Campaigns **target** particular segments — exposure is not random.
`expected_uplift`	float	The increase in purchase probability the campaign designer **expected** (≈0.02–0.14). A planning assumption, **not** a measured result.

campaign_spend_wide.csv

Column	Type	Description
`campaign_id`	int	Campaign identifier. Links to `campaigns.csv`.
`spend_2021`	float	Campaign spend in USD during 2021.
`spend_2022`	float	Campaign spend in USD during 2022.
`spend_2023`	float	Campaign spend in USD during 2023.

> **Note**
>
> This table is stored with one column per year. A missing value means the campaign was not running in that year.

events.csv

Column	Type	Description
`EventID`	int	Unique identifier for the event.
`TimeStamp`	datetime	When the event occurred.
`cust_id`	int	Customer performing the event. Links to the customer files.
`SessionID`	int	Groups events belonging to a single visit. One customer has many sessions.
`Event Type`	string	The action taken: view, click, add_to_cart, bounce, purchase.
`prod_id`	float	Product involved, where applicable. Links to `products.csv`. Missing where the event had no product (e.g. a bounce, or a home-page visit).
`device`	string	Device used: mobile, desktop, tablet. Some values are missing.

Column	Type	Description
`Traffic Source`	string	What brought the user to the site: Organic, Email, Paid Search, Social, Direct.
`campaign_id`	int	Campaign linked to the event. `0` means **no campaign** — it does not correspond to any row in `campaigns.csv`.
`PageCategory`	string	Page where the event occurred: Home, PLP (product list), PDP (product detail), Cart, Checkout.
`session duration (sec)`	float	Length of the session in seconds. **A session lasts minutes, not days.**
`ExperimentGroup`	string	The customer's assigned checkout version, consistent with `customer_profile.csv`.

transactions.csv

Column	Type	Description
`TxnID`	int	Identifier for the purchase.
`TimeStamp`	datetime	When the purchase occurred.
`cust_id`	int	Purchasing customer. Links to the customer files.
`prod_id`	float	Product purchased. Links to `products.csv`. Missing on some rows.
`Qty`	int	Number of units bought. A purchase involves at least one unit.
`discount %`	float	Discount rate applied to the purchase (0–0.20).
`Revenue (USD)`	float	Revenue from the transaction. **Negative for refunds.** Missing on some rows.
`campaign_id`	int	Campaign associated with the purchase. `0` = no campaign.
`Refunded`	string	Whether the transaction was refunded: "Yes" or "No".